Chapter 8. Dirichlet's Theorem

Primes in arithmetic progression via characters

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8.1 A little elementary number theory

Motivation. Dirichlet's theorem on primes in arithmetic progressions is proved by Fourier analysis on the finite abelian group $\mathbb{Z}^*(q)$. This section collects the arithmetic needed: unique factorization, and Euler's analytic proof that there are infinitely many primes via $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ and the divergence of $\sum 1/p$.

8.1.1 The fundamental theorem of arithmetic

For integers a, b, $a \mid b$ means $b = ac$ for some $c \in \mathbb{Z}$. A prime is an integer $p > 1$ whose only positive divisors are 1 and p. The greatest common divisor $\gcd(a, b)$ of positive integers is the largest positive common divisor; a, b are relatively prime if $\gcd(a, b) = 1$.

Theorem 1.1 (Euclid's algorithm)

For any integers a and b with $b > 0$ there exist unique integers q and r with $0 \le r < b$ such that $a = qb + r$.

Proof.

Step Existence. The set $\{a - qb : q \in \mathbb{Z}\} \cap \{0, 1, 2, \dots\}$ is nonempty (take q very negative), so it has a least element $r = a - qb \ge 0$. If $r \ge b$ then $r - b = a - (q+1)b \ge 0$ is smaller, a contradiction; so $0 \le r < b$.

Step Uniqueness. If $a = qb + r = q'b + r'$ with $0 \le r, r' < b$, then $b \mid r - r'$ and $|r - r'| < b$, so $r = r'$ and then $q = q'$. ∎

Theorem 1.2

If $\gcd(a, b) = d$, then there exist integers x and y such that $ax + by = d$.

Proof. Let S be the set of positive integers of the form $ax + by$, and $s = ax + by$ its least element. Every common divisor of a, b divides s, so $d \le s$. Write $a = qs + r$ with $0 \le r < s$; then $r = a(1 - qx) + b(-qy)$ is of the same form, so by minimality $r = 0$, i.e. $s \mid a$. Likewise $s \mid b$, so $s \le d$. Hence $s = d$. ∎

Corollary 1.3

Positive integers a and b are relatively prime if and only if $ax + by = 1$ for some integers x, y.

Proof. One direction is Theorem 1.2. Conversely, if $ax + by = 1$ then any common positive divisor of a, b divides 1. ∎

Corollary 1.4

If a and c are relatively prime and $c \mid ab$, then $c \mid b$. In particular, if a prime p does not divide a and $p \mid ab$, then $p \mid b$.

Proof. Write $1 = ax + cy$; then $b = abx + cby$, and c divides both terms. For a prime $p \nmid a$ we have $\gcd(a, p) = 1$. ∎

Corollary 1.5

If a prime p divides $a_1 \cdots a_r$, then $p \mid a_i$ for some i.

Proof. If $p \nmid a_1$ then $p \mid a_2 \cdots a_r$ by Corollary 1.4; induct on r. ■

Theorem 1.6

Every integer greater than 1 factors uniquely into a product of primes (up to order).

Proof.

Step Existence. Suppose not, and let n be the smallest integer > 1 without a factorization. Then n is not prime, so $n = ab$ with $1 < a, b < n$. Both a and b factor, and hence so does n, a contradiction.

Step Uniqueness. Suppose $n = p_1 \cdots p_r = q_1 \cdots q_s$. By Corollary 1.5, $p_1 \mid q_j$ for some j, so $p_1 = q_j$. Cancel and induct on r. ■

8.1.2 The infinitude of primes

Theorem 1.7

There are infinitely many primes.

Proof. If $p_1, \ldots, p_n$ were all the primes, $N = p_1 \cdots p_n + 1$ would have a prime factor (Theorem 1.6), which must be some p_i. But $p_i \mid N - p_1 \cdots p_n = 1$, which is absurd. ■

Euler's analytic argument proves more ($\sum 1/p = \infty$) and is the model for Dirichlet's proof. For $s > 1$ let $\zeta(s) = \sum_{n \geq 1} n^{-s}$. The series converges, and comparing with $\int x^{-s} dx$,

$$\frac{1}{s-1} \leq \zeta(s) \leq 1 + \frac{1}{s-1}, \qquad \text{so } \zeta(s) \to \infty \text{ as } s \to 1^+. \tag{8.1}$$

(Indeed $\int_n^{n+1} x^{-s} dx \leq n^{-s} \leq \int_{n-1}^{n} x^{-s} dx$ for $n \geq 2$.) An infinite product $\prod A_n$ **converges** if $\prod_{n \leq N} A_n$ has a limit as $N \to \infty$.

Lemma 1.8

The exponential and logarithm functions satisfy:
(i) $e^{\log x} = x$ for $x > 0$.
(ii) $\log(1 + x) = x + E(x)$ with $|E(x)| \leq x^2$ if $|x| < 1/2$.
(iii) If $\log(1 + x) = y$ and $|x| < 1/2$, then $|y| \leq 2|x|$.

Proof. (i) is the definition of log as the inverse of exp.

(ii) For $|x| < 1$, $\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$, so $|E(x)| \leq \frac{x^2}{2}(1 + |x| + |x|^2 + \cdots) = \frac{x^2}{2(1-|x|)} \leq x^2$ when $|x| < 1/2$.

(iii) By (ii), $|y| \leq |x| + x^2 \leq |x| + \frac{1}{2}|x| \leq 2|x|$. ■

Proposition 1.9

If $A_n = 1 + a_n$ and $\sum |a_n|$ converges, then $\prod_n A_n$ converges, and the product vanishes if and only if some factor A_n vanishes. Also, if $a_n \neq 1$ for all n, then $\prod_n 1/(1 - a_n)$ converges.

Proof. Real a_n here (the complex case is Proposition 3.2). Discard the finitely many n with $|a_n| \geq 1/2$; they only multiply the limit by a fixed finite factor (zero exactly when some $A_n =$

0). For the rest, $\prod_{n\le N} A_n = e^{B_N}$ with $B_N = \sum_{n\le N}\log(1+a_n)$, and $|\log(1+a_n)| \le 2|a_n|$ by Lemma 1.8(iii). So $B_N \to B$, and by continuity of exp, $\prod A_n \to e^B \neq 0$. For the last statement, $1/(1-a_n) = 1 + a_n/(1-a_n)$ and $|a_n/(1-a_n)| \le 2|a_n|$ once $|a_n| < 1/2$. ∎

Theorem 1.10

For every $s > 1$, $\zeta(s) = \prod_p \dfrac{1}{1-p^{-s}}$, the product being over all primes.

Proof. The product converges by Proposition 1.9 with $a_n = p_n^{-s}$, since $\sum_p p^{-s} \le \sum n^{-s} < \infty$. Let $p_1 < \cdots < p_N$ be the primes $\le M$, and expand each factor as a geometric series:

$$\prod_{j\le N} \frac{1}{1-p_j^{-s}} = \prod_{j\le N}\left(1 + p_j^{-s} + p_j^{-2s} + \cdots\right) = \sum_{n\in\mathcal{N}_M} n^{-s},$$

where by unique factorization (Theorem 1.6) $\mathcal{N}_M$ is the set of integers all of whose prime factors are $\le M$, each appearing once (absolutely convergent series may be multiplied out and rearranged). Since $\{1,\dots,M\} \subset \mathcal{N}_M$,

$$\sum_{n\le M} n^{-s} \le \prod_{p\le M} \frac{1}{1-p^{-s}} \le \zeta(s).$$

Let $M \to \infty$. ∎

Proposition 1.11

The series $\sum_p 1/p$ over all primes diverges.

Proof. Take logarithms in Theorem 1.10 (legitimate since the partial products converge to a positive limit and log is continuous): by Lemma 1.8(ii),

$$\log\zeta(s) = -\sum_p \log(1-p^{-s}) = \sum_p \left(p^{-s} + O(p^{-2s})\right) = \sum_p p^{-s} + O(1),$$

since $\sum_p p^{-2s} \le \sum n^{-2}$ for $s \ge 1$. As $s \to 1^+$, $\zeta(s) \to \infty$ by (8.1), so $\sum_p p^{-s} \to \infty$. Since $p^{-s} \le p^{-1}$, $\sum_p 1/p = \infty$. ∎

8.2 Dirichlet's theorem

Motivation. To count primes $p \equiv \ell \pmod q$ we must isolate the progression inside $\sum_p p^{-s}$. The indicator δ_ℓ of the class ℓ is a function on the group $\mathbb{Z}^*(q)$, so it expands in characters; each character gives a series $\sum \chi(p)p^{-s}$, and the theorem reduces to showing that these stay bounded for $\chi \neq \chi_0$, i.e. to the behaviour of the L-functions at $s = 1$.

Theorem 2.1 (Dirichlet)

If q and ℓ are relatively prime positive integers, then there are infinitely many primes of the form $\ell + kq$ with $k \in \mathbb{Z}$.

Following Euler, we shall show that $\sum_{p\equiv\ell\,(q)} p^{-s} \to \infty$ as $s \to 1^+$; the proof is completed in Section 8.3.

8.2.1 Fourier analysis, Dirichlet characters, and reduction of the theorem

$\mathbb{Z}^*(q)$ denotes the multiplicative group of residues mod q that are relatively prime to q (a group by Corollary 1.3); its order is Euler's $\varphi(q)$. Let δ_ℓ be the indicator of ℓ on $\mathbb{Z}^*(q)$. Since $\mathbb{Z}^*(q)$ is finite abelian, Theorem 2.7 gives

$$\delta_\ell(m) = \sum_{e \in \widehat{\mathbb{Z}^*(q)}} \hat{\delta}_\ell(e) e(m), \qquad \hat{\delta}_\ell(e) = \frac{1}{\varphi(q)} \sum_{m \in \mathbb{Z}^*(q)} \delta_\ell(m) \overline{e(m)} = \frac{1}{\varphi(q)} \overline{e(\ell)}.$$

For a character e of $\mathbb{Z}^*(q)$, the **Dirichlet character** $\chi = \chi_e$ is the function on $\mathbb{Z}$ with $\chi(m) = e(m \bmod q)$ if $\gcd(m,q) = 1$ and $\chi(m) = 0$ otherwise. The trivial Dirichlet character is χ_0 (from $e \equiv 1$). Extend δ_ℓ to $\mathbb{Z}$ by $\delta_\ell(m) = 1$ if $m \equiv \ell \pmod q$, 0 otherwise.

Lemma 2.2

The Dirichlet characters are multiplicative: $\chi(mn) = \chi(m)\chi(n)$ for all $m, n \in \mathbb{Z}$. Moreover

$$\delta_\ell(m) = \frac{1}{\varphi(q)} \sum_{\chi} \overline{\chi(\ell)} \chi(m),$$

where the sum is over all Dirichlet characters mod q.

Proof. If m, n are both prime to q, so is mn, and $\chi(mn) = e(mn) = e(m)e(n)$. If either is not, then neither is mn, and both sides vanish. The formula holds for m prime to q by the expansion above. If $\gcd(m,q) > 1$, both sides are 0: the left because ℓ is prime to q and so $m \not\equiv \ell$, the right because $\chi(m) = 0$. ∎

Hence, for $s > 1$,

$$\sum_{p \equiv \ell\,(q)} \frac{1}{p^s} = \sum_p \frac{\delta_\ell(p)}{p^s} = \frac{1}{\varphi(q)} \sum_{p \nmid q} \frac{1}{p^s} + \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \overline{\chi(\ell)} \sum_p \frac{\chi(p)}{p^s}. \tag{8.2}$$

The first term on the right tends to ∞ as $s \to 1^+$ (Proposition 1.11; only finitely many primes divide q). So Dirichlet's theorem follows from:

Theorem 2.3

If χ is a nontrivial Dirichlet character, then $\sum_p \chi(p) p^{-s}$ remains bounded as $s \to 1^+$.

The proof occupies Section 8.3.

8.2.2 Dirichlet L-functions

For a Dirichlet character χ and $s > 1$ define

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

Theorem 2.4 (Dirichlet product formula)

If $s > 1$, then

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \frac{1}{1 - \chi(p) p^{-s}}.$$

Proof. Identical to Theorem 1.10, using $|\chi(p)p^{-s}| \le p^{-s}$ (so the product converges by Proposition 3.2) and the multiplicativity of χ (Lemma 2.2): expanding $\prod_{p\le M}(1+\chi(p)p^{-s}+\chi(p)^2p^{-2s}+\cdots)$ gives $\sum_{n\in\mathcal{N}_M}\chi(n)n^{-s}$, since $\chi(p_1^{a_1}\cdots p_k^{a_k}) = \chi(p_1)^{a_1}\cdots\chi(p_k)^{a_k}$. The difference from $\sum_n \chi(n)n^{-s}$ is bounded by $\sum_{n>M} n^{-s} \to 0$. ∎

Formally, taking logarithms and using $\log(1+x) = x + O(x^2)$,

$$\log L(s,\chi) = -\sum_p \log\left(1-\chi(p)p^{-s}\right) = \sum_p \frac{\chi(p)}{p^s} + O(1).$$

So if $L(1,\chi)$ is finite and nonzero, $\sum_p \chi(p)p^{-s}$ is bounded as $s \to 1^+$. Making this rigorous requires:

(a) a logarithm of complex numbers, since χ may be complex-valued;

(b) continuity of $L(s,\chi)$ down to $s=1$; and (c) the key fact $L(1,\chi) \neq 0$.

8.3 Proof of the theorem

Motivation. Three ingredients make the formal argument rigorous: a power-series logarithm $\log_1$ on the disc and a logarithm $\log_2$ of $L(s,\chi)$ defined by integrating L'/L; the extension of $L(s,\chi)$ to $s>0$ via the bounded partial sums of a nontrivial character; and the non-vanishing $L(1,\chi)\neq 0$, proved for complex χ by a positivity argument and for real χ by summation along hyperbolas.

8.3.1 Logarithms

For $|z|<1$ define

$$\log_1\left(\frac{1}{1-z}\right) = \sum_{k=1}^{\infty}\frac{z^k}{k}.$$

Proposition 3.1

The function $\log_1$ satisfies:
(i) If $|z|<1$, then $e^{\log_1(\frac{1}{1-z})} = \frac{1}{1-z}$.
(ii) If $|z|<1$, then $\log_1\left(\frac{1}{1-z}\right) = z + E_1(z)$, where $|E_1(z)| \le |z|^2$ if $|z| < 1/2$.
(iii) If $|z|<1/2$, then $\left|\log_1\left(\frac{1}{1-z}\right)\right| \le 2|z|$.

Proof. (i) Write $z = re^{i\theta}$, $0\le r<1$; it suffices to show $(1-re^{i\theta})\,e^{\sum_k (re^{i\theta})^k/k} = 1$. Differentiating in r gives

$$\Big[-e^{i\theta} + (1-re^{i\theta})\,e^{i\theta}\sum_{k\ge 1}(re^{i\theta})^{k-1}\Big]e^{\sum_k (re^{i\theta})^k/k} = \Big[-e^{i\theta} + (1-re^{i\theta})\frac{e^{i\theta}}{1-re^{i\theta}}\Big]e^{\cdots} = 0.$$

So the left side is constant in r, and at $r=0$ it equals 1.

(ii) $|E_1(z)| = \left|\sum_{k\ge 2} z^k/k\right| \le \frac{|z|^2}{2}\sum_{j\ge 0}|z|^j = \frac{|z|^2}{2(1-|z|)} \le |z|^2$ for $|z|<1/2$.

(iii) By (ii), $\left|\log_1(\frac{1}{1-z})\right| \le |z| + |z|^2 \le 2|z|$. ∎

Proposition 3.2

If $\sum|a_n|$ converges and $a_n \neq 1$ for all n, then $\prod_{n=1}^{\infty}\frac{1}{1-a_n}$ converges and is nonzero.

Proof. As in Proposition 1.9: for $n \geq n_0$ we have $|a_n| < 1/2$, and by Proposition 3.1(i),

$$\prod_{n_0 \leq n \leq N} \frac{1}{1-a_n} = \exp\Big(\sum_{n_0 \leq n \leq N} \log_1 \Big(\frac{1}{1-a_n} \Big) \Big).$$

The series in the exponent converges absolutely by (iii), so the partial products converge to $e^B \neq 0$. The finitely many factors with $n < n_0$ are finite and nonzero. ∎

8.3.2 L-functions

Proposition 3.3

Let χ_0 be the trivial Dirichlet character ($\chi_0(n) = 1$ if $\gcd(n,q) = 1$, 0 otherwise) and $q = p_1^{a_1} \cdots p_N^{a_N}$. Then

$$L(s,\chi_0) = (1-p_1^{-s})(1-p_2^{-s}) \cdots (1-p_N^{-s})\, \zeta(s).$$

Therefore $L(s,\chi_0) \to \infty$ as $s \to 1^+$.

Proof. By Theorems 2.4 and 1.10, $L(s,\chi_0) = \prod_{p \nmid q} (1-p^{-s})^{-1}$ is the Euler product for ζ with the factors for $p_1, \ldots, p_N$ removed. The last statement follows from (8.1). ∎

Proposition 3.4

If χ is a nontrivial Dirichlet character, then $\sum_{n \geq 1} \chi(n)/n^s$ converges for $s > 0$; we denote its sum by $L(s,\chi)$. Moreover:
(i) $L(s,\chi)$ is continuously differentiable for $0 < s < \infty$.
(ii) There are constants $c, c' > 0$ with $L(s,\chi) = 1 + O(e^{-cs})$ and $L'(s,\chi) = O(e^{-c's})$ as $s \to \infty$.

Lemma 3.5

If χ is a nontrivial Dirichlet character, then $\big|\sum_{n=1}^{k} \chi(n)\big| \leq q$ for all k.

Proof. First $S = \sum_{n=1}^{q} \chi(n) = 0$: for $a \in \mathbb{Z}^*(q)$, multiplicativity gives $\chi(a)S = \sum_n \chi(an) = S$, because $n \mapsto an$ permutes the residues mod q; and $\chi(a) \neq 1$ for some a. Since χ is q-periodic, writing $k = aq + b$ with $0 \leq b < q$ leaves $\sum_{n=1}^{k} \chi(n) = \sum_{aq < n \leq aq+b} \chi(n)$, at most q terms each of modulus ≤ 1. ∎

Proof of Proposition 3.4. Let $s_k = \sum_{n=1}^{k} \chi(n)$, $s_0 = 0$, so $|s_k| \leq q$. Summation by parts gives

$$\sum_{n=1}^{N} \frac{\chi(n)}{n^s} = \sum_{n=1}^{N} \frac{s_n - s_{n-1}}{n^s} = \sum_{n=1}^{N-1} s_n \Big[\frac{1}{n^s} - \frac{1}{(n+1)^s} \Big] + \frac{s_N}{N^s}.$$

Let $f(x) = x^{-s}$; by the mean value theorem $|n^{-s} - (n+1)^{-s}| \leq \sup_{[n,n+1]} |f'| \leq s\, n^{-s-1}$. For $s \geq \delta > 0$ the boundary term tends to 0, and the sum converges absolutely and uniformly on compact subsets of $[\delta, \infty)$, since $|s_n|\, s\, n^{-s-1} \leq qs\, n^{-\delta-1}$. So $L(s,\chi) = \sum_n s_n[n^{-s} - (n+1)^{-s}]$ is continuous for $s > 0$.

(i) The term-wise derivative is $\sum_n s_n[-\log n\, n^{-s} + \log(n+1)(n+1)^{-s}]$; with $g(x) = x^{-s} \log x$, $|g'(x)| \leq C_\delta x^{-1-\delta/2}$ for $s \geq \delta$, so this series also converges uniformly on compacts in $(0,\infty)$. Hence $L(\cdot,\chi) \in C^1(0,\infty)$.

(ii) $\chi(1) = 1$, so $|L(s,\chi) - 1| \leq \sum_{n \geq 2} n^{-s} \leq 2^{-s} + \int_2^\infty x^{-s} dx = 2^{-s}(1 + \frac{2}{s-1}) = O(e^{-cs})$ for $s \geq 2$, with any $c < \log 2$. Similarly $|L'(s,\chi)| \leq \sum_{n \geq 2} (\log n) n^{-s} = O(e^{-c's})$. ∎

We now define a logarithm of $L(s,\chi)$ for nontrivial χ and $s>1$ (where $L(s,\chi)\neq 0$ by Theorem 2.4 and Proposition 3.2): since $L(s,\chi)\to 1$ as $s\to\infty$, put

$$\log_2 L(s,\chi) = -\int_s^\infty \frac{L'(t,\chi)}{L(t,\chi)}\,dt,$$

which converges because $L'(t,\chi)/L(t,\chi) = O(e^{-c't})$. (Here $\log_2$ is just a name, not the base-2 logarithm.) If $L(1,\chi)\neq 0$, this integral extends continuously to $s=1$, so $\log_2 L(s,\chi)$ stays bounded as $s\to 1^+$.

Proposition 3.6

If $s>1$, then $e^{\log_2 L(s,\chi)} = L(s,\chi)$. Moreover

$$\log_2 L(s,\chi) = \sum_p \log_1\Big(\frac{1}{1-\chi(p)/p^s}\Big).$$

Proof.

Step First identity. The derivative of $e^{-\log_2 L(s,\chi)}L(s,\chi)$ is $e^{-\log_2 L}(-\frac{L'}{L}L + L') = 0$, so it is constant. As $s\to\infty$, $\log_2 L\to 0$ and $L\to 1$, so the constant is 1.

Step Second identity. Exponentiating the right side and using Proposition 3.1(i) and Theorem 2.4,

$$e^{\sum_p \log_1(\frac{1}{1-\chi(p)p^{-s}})} = \prod_p \frac{1}{1-\chi(p)p^{-s}} = L(s,\chi).$$

So both sides have the same exponential, and their difference is $2\pi i M(s)$ with $M(s)\in\mathbb{Z}$. Both sides are continuous in $s>1$ (the series converges uniformly on $[\delta,\infty)$, $\delta>1$, by (iii) of Proposition 3.1). Hence M is continuous and integer-valued, thus constant. As $s\to\infty$ both sides tend to 0, so $M\equiv 0$. ∎

By Proposition 3.1(ii),

$$\log_2 L(s,\chi) = \sum_p \log_1\Big(\frac{1}{1-\chi(p)/p^s}\Big) = \sum_p \frac{\chi(p)}{p^s} + O\Big(\sum_p \frac{1}{p^{2s}}\Big) = \sum_p \frac{\chi(p)}{p^s} + O(1).$$

So if $L(1,\chi)\neq 0$, then $\log_2 L(s,\chi)$ is bounded as $s\to 1^+$, hence so is $\sum_p \chi(p)p^{-s}$. Theorem 2.3, and with it Dirichlet's theorem, therefore reduces to:

8.3.3 Non-vanishing of the L-function

Theorem 3.7

If $\chi\neq\chi_0$, then $L(1,\chi)\neq 0$.

The proof splits into two cases. A Dirichlet character is **real** if it takes only real values $(0,\pm 1)$ and **complex** otherwise; equivalently, χ is real iff $\chi=\bar\chi$.

Case I: complex Dirichlet characters

Lemma 3.8

If $s>1$, then $\prod_\chi L(s,\chi)\geq 1$, the product being over all Dirichlet characters mod q. In particular the product is real-valued.

Proof. By Proposition 3.6 (and its analogue for χ_0, proved the same way from the Euler product) $L(s,\chi) = \exp\big(\sum_p \log_1 \frac{1}{1-\chi(p)p^{-s}}\big)$. Hence

$$\prod_{\chi} L(s,\chi) = \exp\Big(\sum_{\chi}\sum_{p}\sum_{k\ge 1}\frac{1}{k}\frac{\chi(p^k)}{p^{ks}}\Big) = \exp\Big(\sum_{p}\sum_{k\ge 1}\frac{1}{kp^{ks}}\sum_{\chi}\chi(p^k)\Big),$$

using $\chi(p)^k = \chi(p^k)$; the rearrangement is justified by absolute convergence. By Lemma 2.2 with $\ell = 1$, $\sum_\chi \chi(p^k) = \varphi(q)\delta_1(p^k) \ge 0$. So the exponent is real and non-negative, and the product is ≥ 1. ∎

Lemma 3.9

The following hold:
(i) If $L(1,\chi) = 0$, then $L(1,\bar\chi) = 0$.
(ii) If χ is nontrivial and $L(1,\chi) = 0$, then $|L(s,\chi)| \le C|s-1|$ for $1 \le s \le 2$.
(iii) For the trivial character, $|L(s,\chi_0)| \le \frac{C}{|s-1|}$ for $1 < s \le 2$.

Proof. (i) $L(1,\bar\chi) = \overline{L(1,\chi)}$, since the series has real terms n^{-s} multiplying $\chi(n)$.

(ii) By the mean value theorem, since $L(\cdot,\chi)$ is C^1 on $(0,\infty)$ (Proposition 3.4): $|L(s,\chi) - L(1,\chi)| \le \sup_{[1,2]}|L'| \cdot |s-1|$.

(iii) From Proposition 3.3, $|L(s,\chi_0)| \le \zeta(s) \le 1 + \frac{1}{s-1} \le \frac{2}{s-1}$ by (8.1). ∎

Proof of Theorem 3.7 for complex χ. Suppose $L(1,\chi) = 0$. By Lemma 3.9(i), $L(1,\bar\chi) = 0$ too, and $\bar\chi \ne \chi$. So in $\prod_{\chi'} L(s,\chi')$ at least two factors are $O(|s-1|)$ as $s \to 1^+$ (Lemma 3.9(ii)), the factor for χ_0 is $O(1/|s-1|)$ (Lemma 3.9(iii)), and all other factors are bounded (they are continuous on $[1,2]$). Hence the product is $O(|s-1|) \to 0$, contradicting Lemma 3.8. ∎

Case II: real Dirichlet characters

For real χ only one factor $L(s,\chi)$ could vanish, which the pole of $L(s,\chi_0)$ can cancel; a different idea is needed. Dirichlet's method of *summation along hyperbolas* is first illustrated on the divisor problem.

Sums vs. integrals.

Proposition 3.10

If N is a positive integer, then:
(i) $\displaystyle\sum_{1\le n\le N}\frac{1}{n} = \int_1^N \frac{dx}{x} + O(1) = \log N + O(1).$
(ii) More precisely, there is a real number γ, **Euler's constant**, with $\displaystyle\sum_{1\le n\le N}\frac{1}{n} = \log N + \gamma + O(1/N)$.

Proof. It suffices to prove (ii). Let $\gamma_n = \frac{1}{n} - \int_n^{n+1}\frac{dx}{x}$. Since $1/x$ is decreasing, $0 \le \gamma_n \le \frac{1}{n} - \frac{1}{n+1} \le \frac{1}{n^2}$, so $\sum\gamma_n$ converges to some γ, and $\sum_{n>N}\gamma_n \le \sum_{n>N} n^{-2} \le \int_N^\infty x^{-2}dx = \frac{1}{N}$. Then

$$\sum_{n=1}^{N}\frac{1}{n} - \int_1^N\frac{dx}{x} = \sum_{n=1}^{N}\gamma_n + \int_N^{N+1}\frac{dx}{x} = \gamma - \sum_{n>N}\gamma_n + \int_N^{N+1}\frac{dx}{x},$$

and the last two terms are $O(1/N)$. ∎

Proposition 3.11

If N is a positive integer, then

$$\sum_{1\le n\le N}\frac{1}{n^{1/2}}=\int_1^N\frac{dx}{x^{1/2}}+c'+O(N^{-1/2})=2N^{1/2}+c+O(N^{-1/2}).$$

Proof. [Supplement] Let $\gamma_n=n^{-1/2}-\int_n^{n+1}x^{-1/2}dx$. Then $0\le\gamma_n\le n^{-1/2}-(n+1)^{-1/2}\le\frac{1}{2}n^{-3/2}$ by the mean value theorem for $x^{-1/2}$. So $\sum\gamma_n=c'$ converges, with $\sum_{n>N}\gamma_n\le\frac{1}{2}\int_{N-1}^{\infty}x^{-3/2}dx=O(N^{-1/2})$. As before,

$$\sum_{n\le N}n^{-1/2}-\int_1^N x^{-1/2}dx=c'-\sum_{n>N}\gamma_n+\int_N^{N+1}x^{-1/2}dx=c'+O(N^{-1/2}),$$

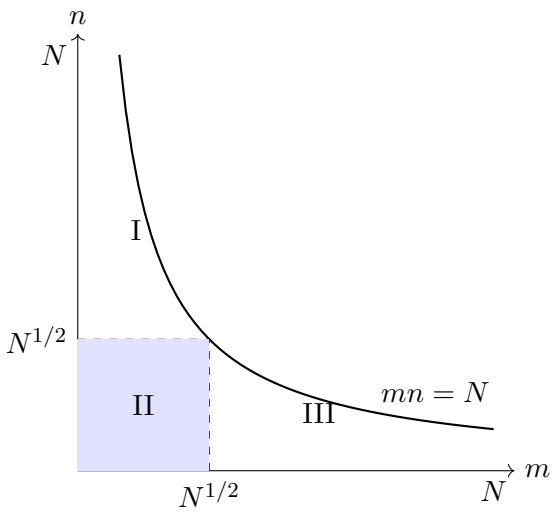
and $\int_1^N x^{-1/2}dx=2N^{1/2}-2$; take $c=c'-2$. ∎

Hyperbolic sums. For F defined on pairs of positive integers let $S_N=\sum\sum_{mn\le N}F(m,n)$. It can be computed in three ways:

(a) along hyperbolas: $S_N=\sum_{1\le k\le N}\left(\sum_{nm=k}F(m,n)\right)$;

(b) vertically: $S_N=\sum_{1\le m\le N}\left(\sum_{1\le n\le N/m}F(m,n)\right)$;

(c) horizontally: $S_N=\sum_{1\le n\le N}\left(\sum_{1\le m\le N/n}F(m,n)\right)$.



Split the region $\{mn\le N\}$ into $\mathrm{I}=\{1\le m<N^{1/2},\ N^{1/2}<n\le N/m\}$, $\mathrm{II}=\{1\le m\le N^{1/2},\ 1\le n\le N^{1/2}\}$, and $\mathrm{III}=\{N^{1/2}<m\le N/n,\ 1\le n<N^{1/2}\}$, with corresponding sums $S_{\mathrm{I}},S_{\mathrm{II}},S_{\mathrm{III}}$.

Let $d(k)$ be the number of positive divisors of k, so $d(k)=\sum_{nm=k}1$.

Theorem 3.12 (Dirichlet's divisor theorem)

For positive integers N,

$$\frac{1}{N}\sum_{k=1}^{N}d(k)=\log N+O(1).$$

More precisely, $\frac{1}{N}\sum_{k=1}^{N}d(k)=\log N+(2\gamma-1)+O(N^{-1/2})$.

Proof. Let $S_N=\sum_{k\le N}d(k)$, the hyperbolic sum of $F\equiv 1$.

Step First estimate. Summing vertically, $S_N = \sum_{m\le N}[N/m] = \sum_{m\le N}(N/m + O(1)) = N\sum_{m\le N}\frac{1}{m} + O(N)$, and Proposition 3.10(i) gives $S_N/N = \log N + O(1)$.

Step Refined estimate. By symmetry $S_{\mathrm{I}} = S_{\mathrm{III}}$, so $S_N = 2(S_{\mathrm{I}} + S_{\mathrm{II}}) - S_{\mathrm{II}}$. Summing vertically and using Proposition 3.10(ii),

$$S_{\mathrm{I}} + S_{\mathrm{II}} = \sum_{m\le N^{1/2}}[N/m] = N\sum_{m\le N^{1/2}}\frac{1}{m} + O(N^{1/2}) = N\log N^{1/2} + N\gamma + O(N^{1/2}).$$

Also $S_{\mathrm{II}} = [N^{1/2}]^2 = N + O(N^{1/2})$. Hence $S_N = N\log N + (2\gamma - 1)N + O(N^{1/2})$; divide by N. ∎

Non-vanishing for real characters. Let χ be a nontrivial real Dirichlet character, $F(m,n) = \frac{\chi(n)}{(nm)^{1/2}}$, and $S_N = \sum\sum_{mn\le N} F(m,n)$.

Proposition 3.13

The following hold:
(i) $S_N \ge c\log N$ for some constant $c > 0$.
(ii) $S_N = 2N^{1/2}L(1,\chi) + O(1)$.

This proves Theorem 3.7 for real χ: if $L(1,\chi) = 0$, then (ii) says $S_N = O(1)$, contradicting (i).

Summing along hyperbolas, $\sum_{nm=k}\frac{\chi(n)}{(nm)^{1/2}} = \frac{1}{k^{1/2}}\sum_{n|k}\chi(n)$. Part (i) follows from:

Lemma 3.14

For every $k \ge 1$, $\sum_{n|k}\chi(n) \ge 0$, and $\sum_{n|k}\chi(n) \ge 1$ if $k = \ell^2$ for some $\ell \in \mathbb{Z}$.

Proof.

Step Prime powers. If $k = p^a$, the divisors are $1, p, \dots, p^a$, and $\sum_{n|k}\chi(n) = 1 + \chi(p) + \chi(p)^2 + \cdots + \chi(p)^a$. Since $\chi(p) \in \{0, 1, -1\}$, this equals $a + 1$ if $\chi(p) = 1$; 1 if $\chi(p) = -1$ and a is even; 0 if $\chi(p) = -1$ and a is odd; and 1 if $\chi(p) = 0$ (i.e. $p \mid q$). It is always ≥ 0, and ≥ 1 when a is even.

Step General k. If $k = p_1^{a_1}\cdots p_N^{a_N}$, every divisor is $p_1^{b_1}\cdots p_N^{b_N}$ with $0 \le b_j \le a_j$, so by multiplicativity

$$\sum_{n|k}\chi(n) = \prod_{j=1}^{N}(1 + \chi(p_j) + \cdots + \chi(p_j^{a_j})).$$

Each factor is ≥ 0; if k is a square all a_j are even and each factor is ≥ 1. ∎

Proof of Proposition 3.13(i). By the lemma, $S_N = \sum_{k\le N} k^{-1/2}\sum_{n|k}\chi(n) \ge \sum_{\ell\le N^{1/2}}\frac{1}{(\ell^2)^{1/2}} = \sum_{\ell\le N^{1/2}}\frac{1}{\ell} \ge c\log N$, by Proposition 3.10(i). ∎

Lemma 3.15

For all integers $0 < a < b$:
(i) $\displaystyle\sum_{n=a}^{b}\frac{\chi(n)}{n^{1/2}} = O(a^{-1/2})$;
(ii) $\displaystyle\sum_{n=a}^{b}\frac{\chi(n)}{n} = O(a^{-1})$.

Proof. Summation by parts as in Proposition 3.4, with $s_n = \sum_{k\le n}\chi(k)$, $|s_n|\le q$ (Lemma 3.5):

$$\sum_{n=a}^{b}\frac{\chi(n)}{n^{1/2}} = \sum_{n=a}^{b-1} s_n\big[n^{-1/2}-(n+1)^{-1/2}\big] + \frac{s_b}{b^{1/2}} - \frac{s_{a-1}}{a^{1/2}},$$
$$\Big|\sum_{n=a}^{b}\frac{\chi(n)}{n^{1/2}}\Big| \le q\sum_{n=a}^{b-1}\big[n^{-1/2}-(n+1)^{-1/2}\big] + \frac{2q}{a^{1/2}} \le \frac{3q}{a^{1/2}},$$

since the sum telescopes. For (ii) the same argument with n^{-1} in place of $n^{-1/2}$ gives the bound $3q/a$. ∎

Proof of Proposition 3.13(ii). Write $S_N = S_{\mathrm{I}} + (S_{\mathrm{II}} + S_{\mathrm{III}})$.

Step Region I, summed vertically.

$$S_{\mathrm{I}} = \sum_{m<N^{1/2}}\frac{1}{m^{1/2}}\Big(\sum_{N^{1/2}<n\le N/m}\frac{\chi(n)}{n^{1/2}}\Big).$$

By Lemma 3.15(i) the inner sum is $O(N^{-1/4})$, and by Proposition 3.11, $\sum_{m<N^{1/2}} m^{-1/2} = O(N^{1/4})$. So $S_{\mathrm{I}} = O(1)$.

Step Regions II and III, summed horizontally.

$$S_{\mathrm{II}}+S_{\mathrm{III}} = \sum_{n\le N^{1/2}}\frac{\chi(n)}{n^{1/2}}\Big(\sum_{m\le N/n}\frac{1}{m^{1/2}}\Big) = \sum_{n\le N^{1/2}}\frac{\chi(n)}{n^{1/2}}\Big(2\frac{N^{1/2}}{n^{1/2}} + c + O\Big(\frac{n^{1/2}}{N^{1/2}}\Big)\Big)$$

by Proposition 3.11. Split into three sums:

$$2N^{1/2}\sum_{n\le N^{1/2}}\frac{\chi(n)}{n} + c\sum_{n\le N^{1/2}}\frac{\chi(n)}{n^{1/2}} + O\Big(\frac{1}{N^{1/2}}\sum_{n\le N^{1/2}}1\Big).$$

The third is $O(1)$. The second is $O(1)$ by Lemma 3.15(i) with $a=1$. For the first, Lemma 3.15(ii) gives $\sum_{n\le N^{1/2}}\frac{\chi(n)}{n} = L(1,\chi) - \sum_{n>N^{1/2}}\frac{\chi(n)}{n} = L(1,\chi) + O(N^{-1/2})$. Altogether $S_N = 2N^{1/2}L(1,\chi)+O(1)$. ∎

This completes the proof of Theorem 3.7, hence of Theorem 2.3, and by (8.2) $\sum_{p\equiv \ell\,(q)} p^{-s}\to\infty$ as $s\to 1^+$: there are infinitely many primes $p\equiv \ell \pmod q$, which is Dirichlet's Theorem 2.1.